The 043 Training Plan

Pretrain over every 043 dataset, fine-tune per target, decide at matched optimizer steps: arms, evaluation, and the grid across four machines

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October 23, 2026

Abstract

Experiment 043 trains in two stages. Stage 1 pretrains one model over every dataset of the 043 query that has a head, with Ohya 2005 morphology read only as a probe of the frozen model; it carries the arms that change the representation: sampler and weighting, the operator residual (plain against ReZero), where the interaction graphs act (a masked layer, a readout gather, or nowhere), background-strain marks, the isolate route and the mark cap. Stage 2 fine-tunes the stage-1 model on one target at a time, Kemmeren 2014 first, and compares it with training from scratch at the same total optimizer steps and at the same fine-tune steps. The case for the split is the overnight round of 2026-10-22: at seed 0 the single joint objective read below the Kemmeren-alone run at matched steps for two of three arms, and 0.208 to 0.230 on the Sameith 2015 doubles against 0.335 alone, with no validation loss logged. Every arm is called by the decision rule of the design review (a paired gain of at least 0.02 in 5 of 6 seeds, with a resampled interval excluding zero). The grid runs smokes on GilaHyper, eight owner-approved jobs on Delta, the six-seed stage-1 grid and the Kemmeren fine-tunes on Polaris, and the fine-tuning grid on Aurora once the trainer runs on Intel XPU, for an estimated 160 Polaris and 117 Aurora node-hours.

Section status: ✓ ready ■ author-reviewed, keep stable ✗ not done, agents may edit freely

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1 Why two stages ✗

What one joint objective measured. The overnight round of 2026-10-22 trained one objective over the nine datasets of `joint_9` and asked of it a per-target question: does the joint run beat a run on Kemmeren 2014 alone at the same number of optimizer steps (design Sec. 5.4)? At seed 0, on the validation fold, the Kemmeren window mean read 0.078 for the records sampler against 0.110 alone, a transfer of −0.032; −0.020 for the pinball arm; and +0.009 for the balanced sampler. On the 72 Sameith 2015 doubles the alone run read 0.335 and the three joint runs 0.208 to 0.230. One seed is below the decision rule's six, and no run of that round logged a validation loss, so the comparison rests on window means and the training loss and cannot be read for overfitting (design Sec. 6).

Why the answer depended on the batch. In a single objective the transfer to a target is decided by how much of each batch the target holds. Under equal weighting the records sampler gives Kemmeren an expected weight of 0.066 and the promoter library 0.573; the balanced sampler gives every dataset 0.111 (design Table 2). The one positive transfer came from the sampler that moved Kemmeren's weight up. Hypothesis (untested): the sign of the overnight transfer follows the target's expected weight, not the value of the shared representation; one seed cannot separate the two.

The same lesson in 040. The environment case study (experiment 040) built its trainer around one target screen, the others entering as auxiliary sources, and its final list of experiments adds a pooled-then-target arm: train on every screen, then drop the auxiliary sources for the last 10 to 20% of epochs. The direction drawn from both experiments is not to gear a trainer to one target, and to "pretrain" over all phenotypes before fine-tuning for those of interest. For 043 the owner stated it on 2026-10-23: "we want to just be able to train models over all data. then we can eval them at the end, but then we can also fine tune on perturbation data to see if it helps with sameith doubles for example."

Two questions, two stages. The joint objective answers two questions at once. Whether one representation carries every readout is a property of the pretrained model, read per dataset at the end of stage 1 (Sec. 2). Whether that representation helps a given target is a property of the starting point, read by fine-tuning from it against training from scratch at matched optimizer steps (Sec. 3). The stage-1 trainer then needs no target, and the target-specific comparison moves to stage 2, where every batch belongs to the target.

The paper's framing. The manuscript's Fig. 3 states that one latent cell embedding generalizes across systems-biology phenotypes (expression, morphology and fitness), and its introduction places the work beside foundation cell models that pretrain on single-cell expression matrices and fine-tune for phenotype, noting that industrial microbes lack such matrices but have diverse perturbation data. Stage 1 is that claim's test on the 043 readouts, with morphology held out of training and read as a probe; stage 2 is the fine-tuning half.

1.1 Symbols added by this plan ✗

Every symbol of the design review (design Sec. 1, Table 1) keeps its meaning. The plan adds the symbols of Table 1, alphabetical by letter name.

Table 1: Symbols added by the plan. Symbols of the design review are not repeated; "Eq." is the equation of this document that defines the symbol.

Symbol	Meaning	Defined in
α	sampler exponent: dataset s is drawn for a batch slot with probability $\pi_s(\alpha) \propto n_s^\alpha$; $\alpha = 0$ is the balanced sampler	Eq. 1
$\mathrm{bg}(b)$, $\mathcal{P}^{\mathrm{bg}}_{\mathrm{bg}(b)}$	background strain of record b (S288C, BY4741, BY4742, FY4, FY5, Y8205); the fixed marks that strain carries relative to S288C	Eq. 4
$\Delta^{\mathrm{ft}}_{\mathrm{tot}}$, Δ^{ft}_{k}	fine-tune gain against training from scratch for $k_1 + k$ steps, and for k steps	Eq. 7
$\mathbf{E}^{\mathrm{nbr}}_{k}$, $\mathbf{E}^{\mathrm{orig}}$	learned vector of an empty relation-k neighborhood; learned table of mark origin (engineered, background)	Eqs. 3, 4
$\bar{h}^{(k)}_{b,i}$	mean perturbed state of the relation-k neighbors of gene i in record b (readout gather)	Eq. 3
ι_t	origin of mark t: 0 engineered, 1 background	Eq. 4
k_1, k, k_{tot}	optimizer steps of stage 1, of stage 2, and their sum	Eq. 6
lr_2, $\mathcal{F}$	stage-2 learning rate; freezing policy (none, trunk frozen, heads only)	Sec. 3.1
$\pi_s(\alpha)$	probability that a batch slot draws dataset s	Eq. 1
ρ_b	agreement of the route-A and route-B predicted profiles of isolate b, centered	Eq. 5
$\theta^{(1)}$	all trainable parameters at the last epoch of a stage-1 run (the window-end checkpoint)	Eq. 6
W_{ga}	projection of the nine gathered neighbor states, $\in \mathbb{R}^{d \times 9d}$	Eq. 3

2 Stage 1: pretrain over every dataset ✗

2.1 The arm `all` ✗

The stage-1 arm `all` trains every dataset of the 043 query that has a head (Table 2). It differs from `joint_9` by Hu 2007, Jackson 2020 and Nadal-Ribelles 2025, which the owner had left out of `joint_9` on 2026-10-21 and which "all data" brings back. Ohya 2005 trains nothing and is read as a probe of the frozen model (Sec. 4.4); Sameith 2015 Dm is split B, read through Kemmeren's dataset token and standardization; Sameith 2015 Sm stays excluded as a copy of Kemmeren's arrays. Boocock 2025 enters with its 393 pooled records; whether the de novo single cells train is decision P1 of the primer, kept open (Sec. 7). The training record count of `all` is fixed by its prepare stage, which has not run. Estimate (untested): `joint_9`'s 50,767 training records plus about 80% of the added records (the share of single-deletion genes outside folds 0 and 1), about 56,500, or 7,063 steps per epoch at $B = 8$ and $k_1 \approx 353{,}150$ steps over 50 epochs.

Table 2: Datasets of the 043 query and their role in the stage-1 arm `all`. Records are those of the raw store (primer Table 1), before the split. "Trainer spec" says whether `train_043.py` reads the dataset at commit `f1cf262d2`; a dataset without one cannot enter `all` until its spec, transform and standardization are written.

Dataset	Role in `all`	Records in the store, n	Head	Trainer spec at `f1cf262d2`	What `all` needs
Kemmeren 2014	trains	1,482	gene	yes	nothing
Sameith 2015 Dm	split B, evaluation only	72	gene	yes	nothing; read through Kemmeren's token and standardization
Sameith 2015 Sm	excluded	82	–	–	nothing; 82 of 82 identical to Kemmeren's arrays
Hu 2007	trains	262	gene	no	spec; signed X score as released, standardized per dataset
Jackson 2020	trains	784	gene	no	spec; target per primer O7 (deletion effect against the HO control of the same environment)
Nadal-Ribelles 2025	trains	6,118	gene	no	spec; sentinels $\lvert \log_2 \mathrm{FC} \rvert > 20$ set missing before standardization (O6)
Messner 2023	trains	4,624	gene	yes	nothing
Zelezniak 2018	trains	97	gene	yes	nothing
Ohya 2005	probe, never trained	4,678	–	–	the probe script on saved checkpoints (Sec. 4.4)
Caudal 2024	trains	943	gene	yes	nothing
Muenzner 2024 protein	trains	795	gene	yes	nothing
Muenzner 2024 turnover	trains	55	gene	yes	nothing
Boocock 2025	trains, pooled records only	393 of 79,516	gene	yes	nothing; de novo cells wait on P1
N'Guessan 2025	trains	2,951	gene	yes	nothing
Vaishnav 2022	trains	71,158	record	yes	nothing

Every dataset keeps its own gene or record head input, dataset token and bias (design Eq. (13), design Eq. (14)); the objective is design Eq. (20) and design Eq. (21) unchanged, and the validation loss is design Eq. (23), which the prepare-experiment gate requires per dataset.

2.2 Sampler and weighting arms ✗

The two overnight samplers are the ends of one family. A batch slot draws dataset s with probability

$$\pi_s(\alpha) = \frac{n_s^\alpha}{\sum_{s' \in \mathcal{S}} n_{s'}^\alpha}, \qquad \alpha \in [0, 1], \tag{1}$$

then a record of s uniformly, with replacement, for the same number of batches per epoch. $\alpha = 0$ is the balanced sampler of design Sec. 4.4; $\alpha = 1$ matches the records sampler in expectation (the records sampler draws without replacement). $\alpha = 0.5$ is the one new point: Vaishnav, with 42,694 training records against Kemmeren's 1,202, is drawn about 6 times as often as Kemmeren rather than 36 times (records) or equally (balanced).

Under the balanced sampler the two weightings of design Eq. (21) give every dataset the same expected weight $1/|\mathcal{S}|$ (design Table 2), so the weighting contrast is degenerate there and lives under the records sampler only. The distinct cells are therefore four: balanced with equal weighting (the stage-1 reference), records with equal weighting, records with record weighting (the canceled run 6151), and $\alpha = 0.5$ with equal weighting. The reference is the balanced sampler because it is the only joint arm whose seed-0 Kemmeren transfer was positive and whose isolate means were highest (design Sec. 6); that is a choice of reference, not a finding, and the contrasts of Sec. 5.4 decide it.

2.3 Operator residual: plain against ReZero ✗

The overnight operator gates its two sublayers with ReZero scalars initialized at zero (design Eq. (9)), so at step 0 it is the identity and the marks reach nothing until ζ_{attn} moves. The owner's objection, 2026-10-23: "it seems to allow models to cheat ... shut off info before dynamics mature". The plan makes the plain residual the main arm, post-normalized like the trunk (design Eq. (5)) and like the manuscript's operator,

$$h'_i = \mathrm{LN}\big(h_i + \mathrm{Drop}(\Delta h_i)\big), \qquad h_i^{\text{pert}} = \mathrm{LN}\big(h'_i + \mathrm{Drop}(\mathrm{FFN}(h'_i))\big), \tag{2}$$

with Δh_i of design Eq. (8). A record without marks attends only to the null key, whose value is zero, so its $\Delta h_i = 0$ and its state passes through the operator's normalizations alone. ReZero (design Eq. (9)) is the control. The control arm logs ζ_{attn} and ζ_{ffn} every 50 steps, which turns the objection into a measurement: how many steps the gates stay near zero, and whether the ReZero arm's Kemmeren and segregant statistics, the ones that depend on marks, trail the plain arm's at the window.

2.4 Where the graphs act ✗

Off-graph attention is not forbidden in any arm: self-attention stays full in every trunk layer except, in one arm, the single masked layer. Three placements are compared.

Mask (reference). The overnight form: in layer $\ell = 2$ head a attends only along graph $k(a)$ (design Eq. (6)); the other three layers are unmasked.

Readout gather. No layer is masked. The gene head reads, beside the gene's own perturbed state, the mean perturbed state of its neighbors in each relation,

$$\bar{h}_{b,i}^{(k)} = \begin{cases} \dfrac{1}{|\mathcal{N}_k(i)|} \displaystyle\sum_{j \in \mathcal{N}_k(i)} h_{b,j}^{\text{pert}} & \text{if } \mathcal{N}_k(i) \neq \varnothing, \\ \mathbf{E}_k^{\text{nbr}} & \text{otherwise,} \end{cases} \qquad \hat{\mathbf{o}}_{b,i} = \mathrm{MLP}_{\text{gene}}\left([h_{b,i}^{\text{pert}};\ W_{\text{ga}}[\bar{h}_{b,i}^{(1)}; \ldots; \bar{h}_{b,i}^{(9)}];\ \mathbf{c}_b]\right) + \mathbf{b}_{s(b)}^{\text{gene}}, \tag{3}$$

with $\mathcal{N}_k(i) = \{j : A_{ij}^{(k)} = 1 \text{ or } A_{ji}^{(k)} = 1,\ j \neq i\}$, the same nine relations as the mask, and $\mathrm{MLP}_{\text{gene}} : \mathbb{R}^{328} \to \mathbb{R}^{128} \to \mathbb{R}^{n_{\text{out}}}$. A gene with no neighbor in a relation reads a learned vector of that relation, never a zero standing in for a mean. The record head (design Eq. (14)) is unchanged.

None. No mask and no gather: the graphs enter nowhere, the control for both.

Not in the grid. A soft form, an additive learned bias on graph edges in place of $-\infty$ off them, sits between mask and none and is left to the owner (decision 13).

2.5 Background-strain marks ✗

Every non-isolate record is read today on the S288C reference, although the strains carry fixed edits relative to it. The plan adds those edits as marks shared by every record of a background,

$$\mathcal{P}_b = \mathcal{P}_{\text{bg}(b)}^{\text{bg}} \cup \mathcal{P}_b^{\text{eng}}, \qquad \mathbf{p}_t \leftarrow \mathbf{p}_t + \mathbf{E}_{\iota_t}^{\text{orig}}, \tag{4}$$

where $\mathcal{P}_b^{\text{eng}}$ are the record's engineered marks of design Eq. (7), $\iota_t = 1$ for a background mark and $\mathbf{E}^{\text{orig}} \in \mathbb{R}^{2\times d}$, so the operator can tell a shared background edit from the perturbation. Isolates under route A carry their own alleles in their context and take no background marks. Table 3 lists what the loaders name.

Two representation choices are open and are decision 12. The mating type differs between backgrounds at one locus, and the change types of design Sec. 3.3 (step 8) have no mating-type change. A complemented auxotrophy carries a chromosomal deletion and a plasmid copy of the same gene, and the precedence rule of step 8 (deletion outranks present) would read the gene as deleted; the plan proposes two marks on that gene, one per change type. The background is nearly constant within a dataset, so the dataset token and bias can absorb its effect; the backgrounds become separable only where a dataset mixes them: Kemmeren's MATa and MATα strains, Muenzner's FY4 reference beside its isolates, Messner's complemented control.

2.6 Isolates: route A, with the route B consistency test ✗

Route A, each isolate as its own reference context (design Sec. 3.1), stays the main arm. Route B reads an isolate on S288C with its variant, loss-of-function and absence genes as marks, a median of 5,038 marks per Caudal record (job

Table 3: Background strains named by the 043 loaders, and their marks over S288C. "Quoted" means the genotype string is quoted in a loader; a genotype not quoted is the standard published one and is pinned with a verbatim quote and sha256 in a background registry before any mark enters a run.

Background	Genotype over S288C	Datasets whose loader names it	Marks, and what is open
BY4741	MATa his3Δ1 leu2Δ0 met15Δ0 ura3Δ0 (quoted, `hu2007.py`)	Hu 2007; Kemmeren 2014 MATa strains; Messner 2023 and Zelezniak 2018, both made prototrophic	four deletion marks and the mating type; the prototrophy of Messner and Zelezniak (pHLUM minichromosome; Messner's control his3Δ::kanMX complemented by HIS3) puts a deletion and a plasmid copy on one gene
BY4742	MATα his3Δ1 leu2Δ0 lys2Δ0 ura3Δ0 (not yet quoted)	Kemmeren 2014 MATα strains (the loader assigns BY4741 or BY4742 per strain); Sameith 2015 singles, and the doubles except the few the SI places in BY4741	four deletion marks and the mating type
FY4, FY5	MATa, MATα, prototrophic (quoted, `jackson2020.py`)	Jackson 2020; the Muenzner 2024 reference record	the mating type only
S288C::ura3	ura3 deletion (the loader's file-level inference from the SI and the Zenodo token)	Vaishnav 2022, every loaded row	one deletion mark; the per-record strain is not released
Y8205	MATα can1Δ::STE2pr-Sp_his5 lyp1Δ::STE3pr-LEU2 his3Δ1 leu2Δ0 ura3Δ0 (not yet quoted)	none of the loaded rows: Vaishnav's Y8205 random-library file is a separate release the loader does not read	three deletion marks and two integrated cassettes as off-graph present marks, registered for later SGA-derived records

6120); it is not implemented in the trainer (design Sec. 7, item 3). In the route-B arm only the three isolate datasets change route. Beside the per-route statistics of design Eq. (24) and design Eq. (25), a label-free readout compares the two routes' predictions on the same held-out isolate, centered on the training-mean profile $\bar{z}_i$ of design Eq. (25),

$$\rho_b = \mathrm{corr}_i\left(\hat{z}^A_{b,i} - \bar{z}_i,\ \hat{z}^B_{b,i} - \bar{z}_i\right), \qquad \text{reported as } \underset{b}{\mathrm{median}}\,\rho_b \text{ beside } \underset{b}{\mathrm{median}}\,r^A_b \text{ and } \underset{b}{\mathrm{median}}\,r^B_b, \tag{5}$$

for routes A and B trained with the same seed. A median ρ_b well above both r_b means the routes learned the same function of the genotype; a ρ_b near or below them means they reach their scores by different functions, and neither route's isolate score then speaks for the representation.

2.7 The mark cap ✗

The overnight runs kept at most 1,024 marks per record (deletion, absence and loss-of-function marks first), a smoke-time cap the design does not have. N'Guessan carries a median of 2,910 marks and route-B isolates about 5,000. The reference keeps 1,024; the arm raises it to 4,096; an uncapped run is first measured for memory and step time on one GilaHyper card (smoke G6) and enters the grid only if it fits a 40 GB A100.

3 Stage 2: fine-tune per target ✗

3.1 The starting point ✗

A fine-tune starts from $\theta^{(1)}$, the parameters of a stage-1 run at its last epoch, which the trainer already checkpoints as the end of the registered window; validation never picks the checkpoint. The target's standardization is the stage-1 record, matched by its sha256, since the target's training rows are the same in both stages. The dataset token, the per-dataset bias and the head carry over; the optimizer state does not: AdamW restarts at learning rate lr_2 with 100 warmup steps, then constant. The main setting is $\mathrm{lr}_2 = 10^{-4}$, a third of the stage-1 rate, with nothing frozen; the freezing policies $\mathcal{F}$ are none, trunk frozen (the token projections and F_θ fixed, the operator and heads training) and heads only. These are choices, not findings; the Aurora block (Sec. 5.5) tests them, and decision 16 holds the outcome.

3.2 The order of targets ✗

1. **Kemmeren 2014 alone**, read on Kemmeren's validation fold and on the 72 Sameith doubles.
2. **The Sameith singles**: the 82 Kemmeren training records whose genes are the parents of the 72 doubles. Sameith 2015 Sm is the same arrays and stays excluded; the singles are Kemmeren's records, pinned to the training folds by

primer decision P2. Fine-tuning on the parents alone asks whether narrowing to them helps the doubles more than the full Kemmeren set.

3. **The segregant panels**, N'Guessan 2025 then Boocock 2025 pooled, read on their own validation folds.

3.3 The matched-steps comparison ✗

For target s, three runs share the seed of the stage-1 run they are compared with,

$$\begin{aligned} \mathrm{ft}(k): &\quad \theta^{(1)} \text{ after } k_1 \text{ stage-1 steps, then } k \text{ steps on } s, \\ \mathrm{alone}(k_{\mathrm{tot}}): &\quad k_{\mathrm{tot}} = k_1 + k \text{ steps on } s \text{ from scratch}, \\ \mathrm{alone}(k): &\quad k \text{ steps on } s \text{ from scratch}, \end{aligned} \tag{6}$$

and the two gains are

$$\Delta^{\mathrm{ft}}_{\mathrm{tot}} = \overline{\mathrm{stat}}_s^{\,\mathrm{ft}(k)} - \overline{\mathrm{stat}}_s^{\,\mathrm{alone}(k_{\mathrm{tot}})}, \qquad \Delta^{\mathrm{ft}}_{k} = \overline{\mathrm{stat}}_s^{\,\mathrm{ft}(k)} - \overline{\mathrm{stat}}_s^{\,\mathrm{alone}(k)}, \tag{7}$$

each a registered-window mean (design Eq. (28)) over its own run's epochs. $\Delta^{\mathrm{ft}}_{\mathrm{tot}}$ is the headline: positive means pretraining bought more than the same optimizer steps spent on the target. $\Delta^{\mathrm{ft}}_{k} > 0$ with $\Delta^{\mathrm{ft}}_{\mathrm{tot}} \le 0$ means the start helps per fine-tune step but not per total step. One stage-1 run serves every target, so charging all of k_1 to each target makes $\Delta^{\mathrm{ft}}_{\mathrm{tot}}$ the conservative comparison. The stage-1 model's own window mean on s, with no fine-tuning, is reported beside both.

The arithmetic for Kemmeren. With 151 steps per epoch at $B = 8$, the main fine-tune is $k = 210 \times 151 = 31{,}710$ steps, evaluated every 5 epochs, so its window (epochs 169 to 210) holds 9 evaluations. With the estimate $k_1 \approx 353{,}150$ of Sec. 2.1, $k_{\mathrm{tot}} \approx 384{,}860$, or 2,549 Kemmeren epochs with the last one cut (`budget.max_steps`); both are fixed once `all` is prepared. The nearest measured point is the overnight alone run at 317,300 steps: 0.110 on Kemmeren, 0.335 on the doubles (seed 0, job 6152). The singles take $k = 1{,}100$ steps (100 epochs of 11 steps); their alone(k_{tot}) control is the Kemmeren one, which has about 9% more steps than $k_1 + 1{,}100$, a difference that favors the control. The segregant panels take $k = 31{,}710$ as well: about 143 epochs of N'Guessan's 1,770 training records and 1,057 of Boocock's 235.

3.4 What "helps the doubles" means in numbers ✗

The statistic is design Eq. (26): the median over the 72 doubles of the per-double Pearson correlation over 6,127 genes, in released units, centered on Kemmeren's training-mean profile. Table 4 gives the points a fine-tuned model is placed against.

Table 4: Reference points for the Sameith 2015 doubles. Statistic: median over 72 doubles of the centered per-double Pearson r over 6,127 genes (design Eq. (26)). Run values are registered-window means, seed 0; the two rungs use no model and no seed.

Reference	Median centered per-double r	Source
Fitted rung $\lambda_1 y_{\mathrm{strong}} + \lambda_2 y_{\mathrm{weak}} + \lambda_0$, leave one out	0.494	job 6060 (design Eq. (27))
Additive null of the measured singles, $y_x + y_{x'}$	0.440	job 6060
Kemmeren alone, 317,300 steps	0.335	job 6152, run `kmlepsr2`
Joint, three arms, 317,300 steps	0.208 to 0.230	jobs 6136, 6137, 6150

Three statements are registered before the first fine-tune, each under the rule of Sec. 4.2 with the interval resampled over the 72 doubles:

1. **Fine-tuning helps the doubles**: $\Delta^{\mathrm{ft}}_{\mathrm{tot}} \ge 0.02$ on the doubles statistic, positive in at least 5 of 6 seeds, interval excluding zero.
2. **The fine-tuned model clears the additive null**: its doubles window mean exceeds 0.440, with the interval of the paired difference excluding zero. The singles are in training, so the null and the model see the same measured parents.
3. **It clears the fitted rung**: the same against 0.494.

The model's own additive prediction, the sum of its predictions for the two singles, is reported beside each, so a gain can be split into better singles and better combination. The doubles are split B and a test set: they are logged during training but enter only these three statements, read once at the end of the campaign; no choice of arm, lr_2, $\mathcal{F}$ or k reads them, all of which are made on Kemmeren's validation fold.

4 Evaluation ✗

4.1 Split, window and seeds ✗

Both stages use `conf/splits/global_split_v1.json` (design Sec. 5.1): fold 0 test, fold 1 validation, the rest training, and the prepare stage drops fold 0 so its labels are never loaded during development. Each run reads its registered window, the last 20% of its own epochs (design Eq. (28)): epochs 41 to 50 of a 50-epoch stage-1 run, epochs 169 to 210 of the main Kemmeren fine-tune. Validation is logged every evaluation and never selects an epoch. Seeds pair across stages: the stage-2 runs of seed j start from the stage-1 run of seed j, and both of their from-scratch controls take seed j.

4.2 The decision rule ✗

An arm is called better than its named control when the mean paired gain in the primary statistic's window mean is at least 0.02, the gain is positive in at least 5 of 6 paired seeds, and the 95% interval of the gain, resampled over held-out strains (over genes for a per-gene statistic, over doubles for the doubles), excludes zero (design Sec. 5.5). The control is named before the first read: the stage-1 reference for every stage-1 contrast, alone(k_{tot}) and alone(k) for a fine-tune. The rule is committed as `conf/decision_rule.yaml` before the Polaris launch; at commit `f1cf262d2` it stands in the primer and the design review only. Eleven stage-1 contrasts are each called under the rule, which does not adjust for their number; the single read of fold 0 at the end of the campaign is the check on the calls that survive.

4.3 Per-dataset statistics, with the fold-1 ladder beside fold 0 ✗

Every stage-1 run reports each dataset's statistic of design Sec. 5.2 on fold 1, against the linear ladder of `results/linear_evals_043.json` on the same fold (Table 5). The ladder was first reported on fold 0; its fold-1 values train on fold 0 as well but score the same held-out units as the runs, so fold 1 is the column a run is read against during development and fold 0 the column it is read against once at the end.

Table 5: Per-dataset statistics and the linear ladder, fold 1 beside fold 0. Held-out units are fold-1 counts (fold 0 within two). Bars are the named linear model of `linear_evals_043.py` (jobs 6059 to 6067). A dash means no bar; "inside zero" marks a bar whose value is within noise of zero on a held-out set of fewer than 20 units.

Dataset	Statistic	Held-out units, n	Linear bar, fold 1	Linear bar, fold 0	Linear model of the bar
Kemmeren 2014	median per-gene Pearson r	140	0.131	0.125	ridge, composite token of the deleted gene
Hu 2007	median per-gene Pearson r	19	−0.006	−0.023	same; inside zero
Jackson 2020	median per-gene Pearson r of the effect against HO	–	–	–	none computed
Nadal-Ribelles 2025, environment 0	median per-gene Pearson r	298	0.076	0.088	ridge, composite token
Nadal-Ribelles 2025, environment 1	median per-gene Pearson r	297	0.062	0.065	ridge, composite token
Messner 2023	median per-gene Pearson r	441	0.082	0.083	ridge, composite token
Zelezniak 2018	median per-gene Pearson r	9	0.022	−0.117	ridge, composite token; inside zero
Caudal 2024	median per-gene Pearson r	189	0.372	0.369	GBLUP, kinship from gene-level calls
Muenzner 2024 protein	median per-gene Pearson r	159	0.471	0.423	GBLUP
Muenzner 2024 turnover	median per-gene Pearson r	11	0.113	−0.025	GBLUP; inside zero
Boocock 2025 pooled	median per-gene Pearson r	79	0.064	0.058	ridge on parent of origin
N'Guessan 2025	median per-gene Pearson r	591	0.122	0.155	ridge on parent of origin
Vaishnav 2022	Pearson r over promoters	14,208	0.885	0.884	ridge, k-mer counts (k 1 to 6)
Sameith 2015 doubles	median centered per-double Pearson r (design Eq. (26))	72	0.440 / 0.494	same set	additive measured / fitted, leave one out
Ohya 2005 (probe)	median over 116 moving features of the out-of-fold Pearson r	4,678	0.121	–	ridge, composite token; 5-fold over genotypes

The centered per-strain Pearson (design Eq. (25)) is reported beside the per-gene statistic for every gene-level dataset, with the raw value and Kemmeren's 0.284 different-mutant floor beside any raw number (primer S4).

4.4 The Ohya probe ✗

Ohya 2005 enters no training. For each of its 4,678 genotypes the frozen model produces the perturbed gene state, the mark summary and the mean perturbed state; ridge regressions from each and from their concatenation predict the 116 moving CalMorph features, 5-fold over genotypes, raw and residualized on growth (primer S6). The statistic is the median over the 116 features of the out-of-fold Pearson correlation, with the subgroup of genotypes whose gene is in fold 0 (459) reported as the strictly inductive read. The bar is the same ridge on the frozen composite token: 0.121 raw, 0.083 residualized, 0.137 on the fold-0-gene subgroup; the 019 trained head read 0.149 ± 0.022. The probe is not wired into the trainer; it runs as a CPU job on the window-end checkpoints of every stage-1 and stage-2 run, so it costs no card time and reads how much morphology the representation carries before and after fine-tuning.

4.5 The route consistency readout ✗

For each seed of the route-B arm and the stage-1 reference of the same seed, Eq. 5 is computed on the fold-1 isolates of Caudal and of Muenzner protein and reported as three medians: ρ_b, r_b^A and r_b^B. The readout decides nothing alone; it qualifies the route contrast, since a route-B gain with a low ρ_b is a different function of the genotype rather than a better reading of the same one.

5 The grid ✗

5.1 The cost model ✗

Every hour in this section is an estimate from four measured runs and a machine factor (Table 6). A run's estimate is its optimizer steps times the measured seconds per step of its closest overnight run, times the factor of the machine. The seconds per step are whole-job elapsed time over steps, so they include evaluation and startup. The factors are planning values: A100 at 1.0, A40 at 1.5 and an Aurora tile at 2.0 times the RTX 6000 Ada time. None is measured for this trainer; the first smoke on each machine replaces its factor, and every `--time` and PBS walltime below is recomputed from that measurement before submission. Requested time is about 1.5 times the estimate (the prepare-experiment gate's margin). Route B and the 4,096-mark cap are budgeted at twice the reference's time, a guess that smoke G6 replaces.

Table 6: Measured seconds per optimizer step and the planning factors. Top: GilaHyper, one RTX 6000 Ada card, batch 8, elapsed time over 317,300 steps. Bottom: factors on that time, unmeasured.

Run	Slurm job	Elapsed, h:mm	Seconds per step, s	Used for
`joint_9`, records sampler, HL-Gauss	6136	3:07	0.0353	records-sampler arms
`joint_9`, records sampler, pinball	6150	3:04	0.0348	(agrees with 6136)
`joint_9`, balanced sampler, HL-Gauss	6137	7:13	0.0818	every other stage-1 arm; segregant fine-tunes
Kemmeren alone	6152	2:34	0.0291	Kemmeren fine-tunes and controls
Machine factor, A100 40 GB (Polaris; Delta A100)	–	–	×1.0	planning value
Machine factor, A40 48 GB (Delta)	–	–	×1.5	planning value
Machine factor, Max 1550 tile (Aurora)	–	–	×2.0	planning value

With $k_1 \approx 353{,}150$, a stage-1 reference run is $353{,}150 \times 0.0818\,\text{s} = 8.0\,\text{h}$ on an Ada card, a records-sampler run 3.5 h; a Kemmeren fine-tune of 31,710 steps is 0.26 h; a Kemmeren alone(k_{tot}) of 384,860 steps is 3.1 h; a segregant fine-tune 0.72 h and a segregant alone(k_{tot}) 8.75 h.

5.2 (a) GilaHyper smokes, today ✗

Two cards, one smoke per card at a time, chained `afterany`. Each smoke runs on the real prepared store with the real logger and must pass the gate of Sec. 6 before any run it enables is submitted. Every smoke but G7 waits on code that is not in the trainer at commit `f1cf262d2`; the last column names it.

G2 loads a `joint_9` checkpoint into an `all` model on purpose: the two have different dataset vocabularies, so the smoke proves that rows of $\mathbf{E}^{\text{data}}$ and $\mathbf{b}_s$ are matched by dataset name, not by position.

Table 7: Block (a): GilaHyper smokes. One RTX 6000 Ada card per job, seed 0, batch 8. Estimated hours from Table 6; requested is the slurm `--time`.

Id	Run	Length	Estimated, card-h	Requested, h:mm	Enables	Needs first
G1	`all`, reference arm (balanced, HL-Gauss, equal, plain residual, mask)	2 epochs	0.4	1:00	P0, the Delta request, the Polaris stage-1 job	specs for Hu, Jackson, Nadal-Ribelles; `conf/arm/all.yaml`; prepare of `all`; residual switch
G2	fine-tune path: 6136's window-end checkpoint (`joint_9`) onto Kemmeren	2 epochs (302 steps)	0.1	0:30	D5, D6, Polaris stage 2	checkpoint load mapping dataset rows by name; lr_2, $\mathcal{F}$; standardization sha256 check
G3	readout gather (Eq. 3)	1 epoch	0.2	0:45	C5	graph-placement switch
G4	no graph	1 epoch	0.2	0:45	C6	graph-placement switch
G5	background marks (Eq. 4)	1 epoch	0.2	0:45	C7	background registry, marks, $\mathbf{E}^{\mathrm{orig}}$
G6	memory probe: route B, and uncapped marks, on the largest-mark batches	50 steps each	0.1	0:30	C8, C9, and the factor of 2	route B in the trainer
G7	Kemmeren alone(k), seeds 0 to 5	31,710 steps each	1.6	3:00 (6 jobs of 0:30)	the Δ_k^{ft} controls	nothing
Block total			2.8	7:15		

5.3 (b) Delta, eight individual jobs (owner go required) ✗

Eight single-card jobs on Delta's A40 partition, never the preempt partition, submitted only after the owner's go. Their role is an early read of the whole two-stage chain at two seeds while the Polaris job queues, and a cross-machine check: seeds 1 and 2 repeat inside the Polaris grid. D7 runs first and doubles as the A40 timing probe; D1 to D4 are submitted after its first evaluation, with `--time` recomputed from its measured seconds per step.

Table 8: Block (b): Delta, eight jobs on single A40 cards. Stage-1 runs use the reference configuration except where the arm says otherwise. D5 and D6 depend on D1 and D2 with `afterany` and raise if the checkpoint is missing.

Id	Arm	Seed	Length	Cards, n	Estimated, card-h	Requested, h:mm	Dependency
D1	stage 1, reference (plain residual)	1	50 epochs	1	12.0	18:00	after D7 read
D2	stage 1, reference (plain residual)	2	50 epochs	1	12.0	18:00	after D7 read
D3	stage 1, ReZero control (C1)	1	50 epochs	1	12.0	18:00	after D7 read
D4	stage 1, ReZero control (C1)	2	50 epochs	1	12.0	18:00	after D7 read
D5	stage 2, ft(k) on Kemmeren from D1	1	210 epochs	1	0.4	1:00	afterany D1
D6	stage 2, ft(k) on Kemmeren from D2	2	210 epochs	1	0.4	1:00	afterany D2
D7	Kemmeren alone(k_{tot}), 384,860 steps	1	2,549 epochs	1	4.7	7:00	none
D8	Kemmeren alone(k_{tot}), 384,860 steps	2	2,549 epochs	1	4.7	7:00	none
Block total				8	58.2	88:00	

The alone(k) controls of seeds 1 and 2 come from G7. Two seeds decide nothing under the rule; the block's outputs are the A40 factor, a working chain, and the first paired values of $\Delta_{\mathrm{tot}}^{\mathrm{ft}}$ and Δ_k^{ft}.

5.4 (c) Polaris ✗

Polaris holds the six-seed stage-1 grid. Each node has four A100 40 GB cards and each run takes one card, so a packed job runs four runs per node: one process per card (`CUDA_VISIBLE_DEVICES`), drawing from a task list ordered longest first, each task its own W&B run, a failed task leaving the others running. The queue is `capacity` (1 to 4 nodes, 168 h, one job per project at a time). The `preemptable` queue is not used: the trainer has no resume, so a preemption loses the run. The 1-hour `debug` queue takes the timing probe and the short fine-tunes.

Table 9: Block (c), the Polaris stage-1 job: arms, seeds 0 to 5. One change from the reference per contrast. Estimated hours per run on one A100 card; card-hours are runs times hours.

Id	Arm (change from the reference)	Runs, n	Estimated per run, card-h	Estimated, card-h	Control
R	reference: `all`, balanced, HL-Gauss, equal, plain residual, mask at layer 2, no background marks, route A, cap 1,024	6	8.0	48.0	–
C1	ReZero residual (design Eq. (9))	6	8.0	48.0	R
C2	records sampler, equal weighting	6	3.5	21.0	R
C3	records sampler, record weighting	6	3.5	21.0	C2
C4	sampler $\alpha = 0.5$ (Eq. 1), equal weighting	6	8.0	48.0	R
C5	readout gather, no mask (Eq. 3)	6	8.0	48.0	R
C6	no graph	6	8.0	48.0	R
C7	background marks (Eq. 4)	6	8.0	48.0	R
C8	route B for the isolates, uncapped	6	16.0	96.0	R
C9	mark cap 4,096	6	16.0	96.0	R
C10	pinball loss	6	8.0	48.0	R
C11	`joint_9` at k_1 steps (without Hu, Jackson, Nadal-Ribelles)	6	8.0	48.0	R
K	Kemmeren alone(k_{tot}), the stage-2 control	6	3.1	18.6	–
	Job total	78		636.6	

On 16 cards the job's 636.6 card-hours take about 39.8 h, which is 159.2 node-hours; the longest single runs (16 h) fit well inside that. Every stage-1 run saves its window-end checkpoint, so stage 2 can start from whichever arm the rule names without rerunning stage 1.

Table 10: Block (c): Polaris jobs and node-hours. Estimated node-hours are nodes times the estimated wall time; requested is nodes times the PBS walltime. The allocation is 3,884 node-hours on `GenesisHackathonOct26`.

Id	Job	Queue	Nodes, n	Walltime requested, h	Estimated, node-h	Requested, node-h
P0	timing probe: R for 2 epochs on four cards (seeds 0 to 3), C8 memory, the fine-tune path	debug	1	1	0.5	1
P1	stage-1 grid, Table 9	capacity	4	60	159.2	240
P2	stage 2 from R, seeds 0 to 5: Kemmeren ft(k); singles ft(1,100) and alone(1,100)	debug	1	1	0.6	1
	Block total				160.3	242

5.5 (d) Aurora ✗

Aurora runs only after the trainer runs on Intel XPU and its results agree with Polaris'. Its torch 2.11 XPU environment is live and PyG works on XPU (17 of 17 core checks, job 8938797); what the 043 trainer still needs:

1. **Device.** `TrainConfig043.device` admits `xpu`; device selection, `pin_memory` and the bf16 autocast (`device_type="xpu"`) follow it. At commit `f1cf262d2` the trainer knows only `cuda` and `cpu`.
2. **The `torch_scatter` import chain.** `cell_model_043.py` imports `equivariant_cell_graph_transformer.py`, whose package import reaches `torchcell/models/__init__.py` and `deep_set.py`, which imports `torch_scatter`; there is no XPU build of it. The candidate fix, `torch_geometric.utils.scatter` or a lazy import, is untested.
3. **Attention kernels.** The masked trunk layer and the operator call `scaled_dot_product_attention` with masks; its output on XPU is compared with CUDA on a fixed batch and fixed weights at bf16 tolerance, and its speed is measured.
4. **The launcher.** A PBS script with `ZE_FLAT_DEVICE_HIERARCHY=FLAT`, one process per tile pinned by `ZE_AFFINITY_MASK`, twelve per node, and W&B online through the ALCF proxy.
5. **Data.** The prepared store, the frozen vector tables, the split file and the standardization records on `/flare`, each verified by sha256 against the GilaHyper copy (the transfer starts 2026-10-23).
6. **Parity.** Job A1 runs R and C1, seeds 0 to 5, on twelve tiles. Aurora results enter a decision only if, per dataset of Table 5, the paired Aurora-minus-Polaris window-mean difference has a 95% interval covering zero and a mean below 0.01 in magnitude, half the rule's margin (a proposed threshold).

Table 11: Block (d): Aurora jobs and node-hours. Twelve tiles per node, one run per tile. Estimated tile-hours per run: stage-1 reference 16.0; Kemmeren fine-tune 0.5; segregant fine-tune and alone(k) 1.4; segregant alone(k_{tot}) 17.5. The allocation is 18,468 node-hours.

Id	Job	Queue	Nodes, n	Walltime requested, h	Estimated, node-h	Requested, node-h
A0	XPU port tests: CUDA parity of a fixed batch, one-tile smoke, twelve-tile launcher (4 jobs)	debug	1	1 each	2	4
A1	parity: R and C1, seeds 0 to 5, 50 epochs	capacity	1	24	16	24
A2	stage-2 grid from Polaris R checkpoints: four targets $\times$ $\text{lr}_2 \in \{3, 1, 0.3\} \times 10^{-4}$ $\times$ three policies $\mathcal{F}$ $\times$ 6 seeds (216 runs), plus the segregant and singles controls (30 runs); 406 tile-h	capacity	2	27	35	54
A3	longer stage 1: R and C1 at 200 epochs, seeds 0 to 5 (owner decision 19)	capacity	1	96	64	96
	Block total				117	178

A2 compares its Kemmeren fine-tunes with the Polaris Kemmeren controls (K and G7), a cross-machine comparison admitted only after A1 passes. On two nodes the twelve segregant alone(k_{tot}) runs (17.5 h each) occupy twelve tiles, and the other twelve tiles finish the remaining 196 tile-hours in about 16.3 h, so the wall is about 17.5 h.

5.6 Totals and the order of unblocking ✗

Table 12: Budget per machine. GilaHyper and Delta in card-hours, Polaris and Aurora in node-hours, the unit each allocation is charged in. Share is requested over the allocation.

Machine	Unit	Estimated	Requested	Allocation	Share of allocation, %	Gate
GilaHyper	card-h	2.8	7.3	two cards, owner's grant	–	none
Delta	card-h	58.2	88.0	owner's account	–	owner's go
Polaris	node-h	160.3	242	3,884	6.2	G1, P0
Aurora	node-h	117	178	18,468	1.0	A0, A1

1. **Code**, by the trainer's owner: the three dataset specs, the residual, graph and background switches, the fine-tune entry point, route B, `conf/decision_rule.yaml`, the Ohya probe script, the XPU items of Sec. 5.5.
2. **GilaHyper G1 to G7** (Table 7). G1 passing the gate unblocks P0 and the Delta request; G2 unblocks every fine-tune; G6 fixes the budget of C8 and C9.
3. **Polaris P0** replaces the A100 factor and sets P1's walltime; **P1** then runs the stage-1 grid and the Kemmeren alone(k_{tot}) controls.
4. **Delta**, after the owner's go and in parallel with P1: D7 and D8 first, D1 to D4 on D7's measured factor, D5 and D6 behind D1 and D2.
5. **Aurora A0 then A1**, in parallel with P1; A1 compares with P1's R and C1.
6. **P1's result under the rule** names the stage-1 arm stage 2 starts from (R unless a contrast wins). **P2** runs from its checkpoints; **A2**, if A1 passed, runs the fine-tuning grid; **A3** waits for the owner.
7. **End of campaign**: fold 0 and the three doubles statements of Sec. 3.4 are read once.

6 Launch discipline ✗

The gate. Every machine's first submission is a smoke, never a chain: G1 on GilaHyper, P0 on Polaris, A0 and A1 on Aurora, D7 on Delta. After a smoke has logged two evaluations, `scripts/check_wandb_run_keys.py` is run on its W&B run with `--require` for `val/loss` and, for every dataset of the arm, `val/<dataset>/loss` and its validation score; exit 1 means the trainer is fixed and the smoke rerun before anything depends on it. The gate is rerun after every trainer change, and its output (key, row count, verdict) goes verbatim into the round's dated section of the experiment note.

The logging contract. At every evaluation, as curves: `epoch`, `train/loss` and `val/loss` under the same weighting, the per-dataset pair `train/<dataset>/loss` and `val/<dataset>/loss`, `train/lr`, `train/grad_norm`, every validation statistic of Table 5, and in the ReZero arm ζ_{attn} and ζ_{ffn}. A stage-2 run also logs `stage`, the stage-1 run id it started from and the checkpoint's sha256.

Time. `--time` and PBS walltime are about 1.5 times the estimate from a measured step time on that machine (Sec. 5.1), never a round number chosen in advance. A step-matched run sets `budget.max_steps`, and the trainer raises rather than train a shorter unmatched control.

Chains. Slurm chains use `--dependency=afterany`, PBS chains `-W depend=afterany:<id>`, so a crash never stalls a chain; a fine-tune whose checkpoint is missing raises at startup. GilaHyper uses at most the two cards granted. Delta never uses the preempt partition and never launches without the owner's go. Polaris uses no `preemptable` queue while the trainer has no resume.

W&B. One project, `torchcell_043-expression-proteome-morphology`. Groups are named `s1_<arm>_<loss>_<weighting>` and `s2_<target>_<from-arm>_<lr>_<policy>`, runs `<group>_s<seed>`, with the machine as a tag. `wandb_view_043.py` writes the round's view before the first result is read, loss first (train and validation on one panel, grouped by arm, smoothing on, with a step-axis twin where stage-2 runs and their controls differ in epochs per step), then the headline validation statistics, the per-dataset panels, the train side and bookkeeping. Aborted runs are tagged, not deleted. Links go one per line.

What is committed where. Configs (`conf/arm/all.yaml`, `conf/stage2/*.yaml`, `conf/decision_rule.yaml`), launchers (`scripts/gh_*.slurm`, `scripts/delta_*.slurm`, `scripts/polaris_*.pbs`, `scripts/aurora_*.pbs`) and the task lists of the packed jobs live in `experiments/043-expression-proteome-morphology/` and are committed before submission. Window means, ladder reads and probe reads are JSON files under `results/`, each written by a committed script from the unsampled W&B history or the checkpoints. Each round gets a dated section in the experiment note with the gate output, the queue table (job id, run, dependency, time requested, expected end) and the arms dropped for budget, by name. Checkpoints, prepared stores and job logs stay under `$DATA_ROOT` and the `/eagle` and `/flare` project trees, never in git. The numbers of this document are hand-sourced with `%% SOURCE` comments; a table script in the experiment folder replaces them when the first stage-1 results arrive.

7 Decisions the owner holds ✗

Each item names the choice and the test that settles it. Items 1 to 11 are the open questions of design Sec. 7, shortened, with where the plan places their test; items 12 to 21 are new.

1. **Heterozygous IUPAC codes: typed gap or the encoder's unknown token.** *Test:* re-embed the gapped rows through the unknown token and compare the two arms on Caudal and Muenzner per-gene Pearson, six seeds. Not in the grid; a candidate Aurora stage-1 contrast.
2. **Soft-masked bases.** *Test:* count and re-embed the windows an uppercase slice recovers, then the recovered against the strict-gap arm on the isolate statistics. Not in the grid.
3. **Route B and the consistency test.** *Test:* C8 against R, six seeds, with the consistency readout of Eq. 5.
4. **Attended-token head or head-only tokens.** *Test:* the attended arm against head-only on Kemmeren, the doubles and the isolates, six seeds; not implemented, not in the grid.
5. **Flat-type and six-heads controls.** *Test:* `exp_type_mode: flat` and six heads against the type path on every dataset; the flat control is a config switch and fits a later Polaris job.
6. **Dataset weighting extremes.** *Test:* C2, C3 and C4 against R (Sec. 2.2); under the balanced sampler the weighting contrast is degenerate.
7. **A clade-disjoint isolate split.** *Test:* a second fold file disjoint by Peter 2018 clade; GBLUP and the stage-1 reference on both, the drop from random to clade split being the quantity of interest.
8. **The Addgene 127546 window gate.** *Test:* deposit the plasmid's GenBank file by the manual recipe, compare its window with the reconstruction base by base, re-embed only on a difference; until then, whether Vaishnav numbers may be quoted.
9. **Retiring Sameith 2015 Sm.** *Test:* repeat the identity check on the served graph's records and record it with the retirement decision.

10. **The 188 dropped records.** *Test:* re-query with the `SIZE = 0 OR ANY` shape and confirm the store gains exactly 188 records; `all` would then hold them. Hu, Nadal-Ribelles and Ohya account for 111 of the 188.

11. **The mark cap.** *Test:* G6 for memory, then C9 against R on the segregant statistics.

12. **Background marks.** On or off, how the mating type enters, and whether a complemented auxotrophy is two marks on one gene or one under the precedence rule (Sec. 2.5). *Test:* C7 against R, six seeds, per dataset; and, within Kemmeren, the MATa and MATα strata read separately, since only a mixed-background dataset can separate the marks from the dataset token.

13. **Graph placement.** Mask at one layer, readout gather or none, and whether the soft edge bias is worth an arm. *Test:* C5 and C6 against R, six seeds, on every dataset.

14. **Operator residual.** Plain is the main arm by the owner's preference. *Test:* C1 against R, with the ζ trajectories; if ReZero wins under the rule, whether it replaces plain or stays a control is the owner's call.

15. **The stage-1 reference sampler.** Balanced, records or $\alpha = 0.5$. *Test:* C2 and C4 against R. The winner becomes the reference for any later stage-1 job; stage 2 starts from the winner's checkpoints.

16. **Fine-tune learning rate and freezing policy.** Main $\mathrm{lr}_2 = 10^{-4}$, nothing frozen. *Test:* A2's three rates by three policies, chosen on Kemmeren's validation fold only, never on the doubles.

17. **Fine-tune length.** k = 31,710 for Kemmeren and the segregants, 1,100 for the singles. *Test:* k = 7,550 (50 Kemmeren epochs) beside 31,710 for the main setting, six seeds, Δ_k^{ft} at each; one extra short run per seed.

18. **Which datasets fine-tune second.** The Sameith singles, then the segregant panels, is the proposed order; the isolates (Caudal, Muenzner) are the alternative second target. *Test:* A2 reads every target's $\Delta_{\mathrm{tot}}^{\mathrm{ft}}$ on the same stage-1 checkpoints; an isolate target (Caudal and Muenzner) adds its fine-tunes and 24 control runs, about 230 tile-hours at the segregant rate (an estimate).

19. **A longer stage 1.** Whether 200 stage-1 epochs make a better starting point than 50. *Test:* A3 against R at matched seeds, read on stage 1 and through the Kemmeren fine-tune; 64 node-hours.

20. **Hu, Jackson, Nadal-Ribelles in `all`, and the Boocock de novo cells (P1).** The three were left out of `joint_9` on 2026-10-21; "all data" brings them back. *Test:* C11 (`joint_9` at k_1 steps) against R on the datasets the two share. The de novo cells, 79,123 records beyond the pooled 393, would make Boocock the largest gene-level dataset; their arm waits on P1.

21. **A leave-target-out pretraining control.** `all` minus Kemmeren in stage 1, then the Kemmeren fine-tune, separates transfer from the other datasets from a second pass over the target. *Test:* six seeds of that stage-1 arm and its fine-tunes against R's; about 50 Polaris card-hours, not in the budget.